GPU BUILDER · CUSTOM MEMORY GUIDE

My custom graphics card

A sourcing and placement plan for a 512 GB card built from 32 GB packages: 16 packages in 2 banks on a 512-bit interface.

GENERATED	TARGET	PACKAGES	PLACED
October 24, 2026	512 GB	16 × 32 GB	16 of 16

Before you source

- Memory packages are soldered during board assembly, not socketed; this plan fixes the capacity, bank grouping and interface you would specify.
- Packages must match the GPU's memory interface, voltage and training requirements. Mixed parts on one slice are not a drop-in swap.
- Package density is device-specific: confirm which package sizes the GPU's memory controllers support before committing to a target.

Package counts, bank grouping and bandwidth are teaching estimates computed from the target and module size. This is not a vendor design, a compatibility promise or a bill of materials you can order from a distributor.

The capacity maths

Every number below is derived from the two choices you made, so the plan can be checked with a calculator.

PACKAGES	BANKS	MEMORY BUS	AT 16 Gb/s PER PIN
512 ÷ 32 = 16	⌈16 ÷ 8⌉ = 2	16 × 32 = 512 bits	512 × 16 ÷ 8 = 1024 GB/s theoretical

Bandwidth assumes the teaching model's 16 Gb/s per pin and a fully populated interface. Real cards trade pin rate, package count and power against each other, and application throughput is always lower.

Parts to source

Memory for 16 of 16 packages is marked as placed. Banks are listed as they appear on the board.

The board, power delivery, cooling and chassis hardware are covered by the teaching build guide, not repeated here.

REF.	PART	QTY	SOURCE	WHAT TO BUY
U1	GPU package	1	INTEGRATED	Packaged GPU with a 512-bit memory interface. Not sold as a loose part.
✓ M1–M8	Memory package bank 01	8	INTEGRATED	32 GB GDDR6, x32 interface · Placed. Sold to board makers only; a donor card is the practical source.
✓ M9–M16	Memory package bank 02	8	INTEGRATED	32 GB GDDR6, x32 interface · Placed. Sold to board makers only; a donor card is the practical source.

Placement plan

Work one bank at a time. Each bank is a slice of the memory interface; the last bank may be partial.

01 Fit memory bank 01

Placed

MEMORY · M1–M8

Place M1, M2, M3, M4, M5, M6, M7, M8 around the GPU package, one package per memory channel position.

Where it goes	Bank 01 carries 256 bits of the memory interface (8 × 32-bit packages).
What it holds	8 × 32 GB = 256 GB of the 512 GB target.
Watch for	Keep the packages on the same interface slice and match trace lengths within the slice. A real board also needs a ground reference and decoupling at each package.

Check your maths: 8 × 32 = 256 bits, and 8 × 32 GB = 256 GB of the 512 GB target.

02 Fit memory bank 02

Placed

MEMORY · M9–M16

Place M9, M10, M11, M12, M13, M14, M15, M16 around the GPU package, one package per memory channel position.

Where it goes	Bank 02 carries 256 bits of the memory interface (8 × 32-bit packages).
What it holds	8 × 32 GB = 256 GB of the 512 GB target.
Watch for	Keep the packages on the same interface slice and match trace lengths within the slice. A real board also needs a ground reference and decoupling at each package.

Check your maths: 8 × 32 = 256 bits, and 8 × 32 GB = 256 GB of the 512 GB target.